

In partnership with:  Microsoft

Reactor



.NET Conf 2022

Student Zone

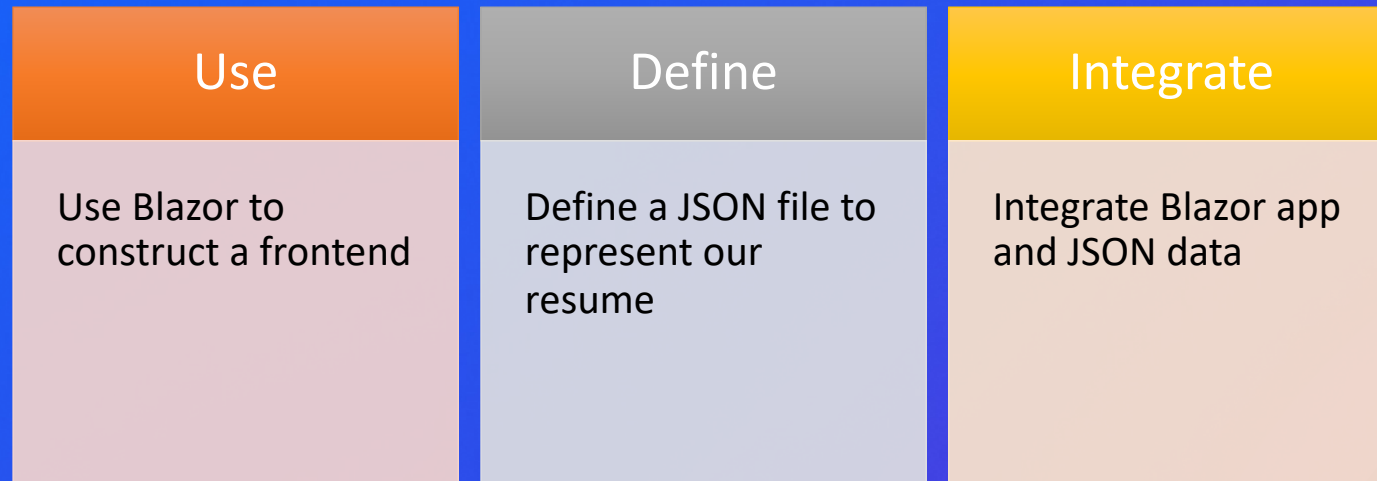


Build a powerful resume app using Blazor

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Our plan

- Build a resume using Blazor



Thinking in Components

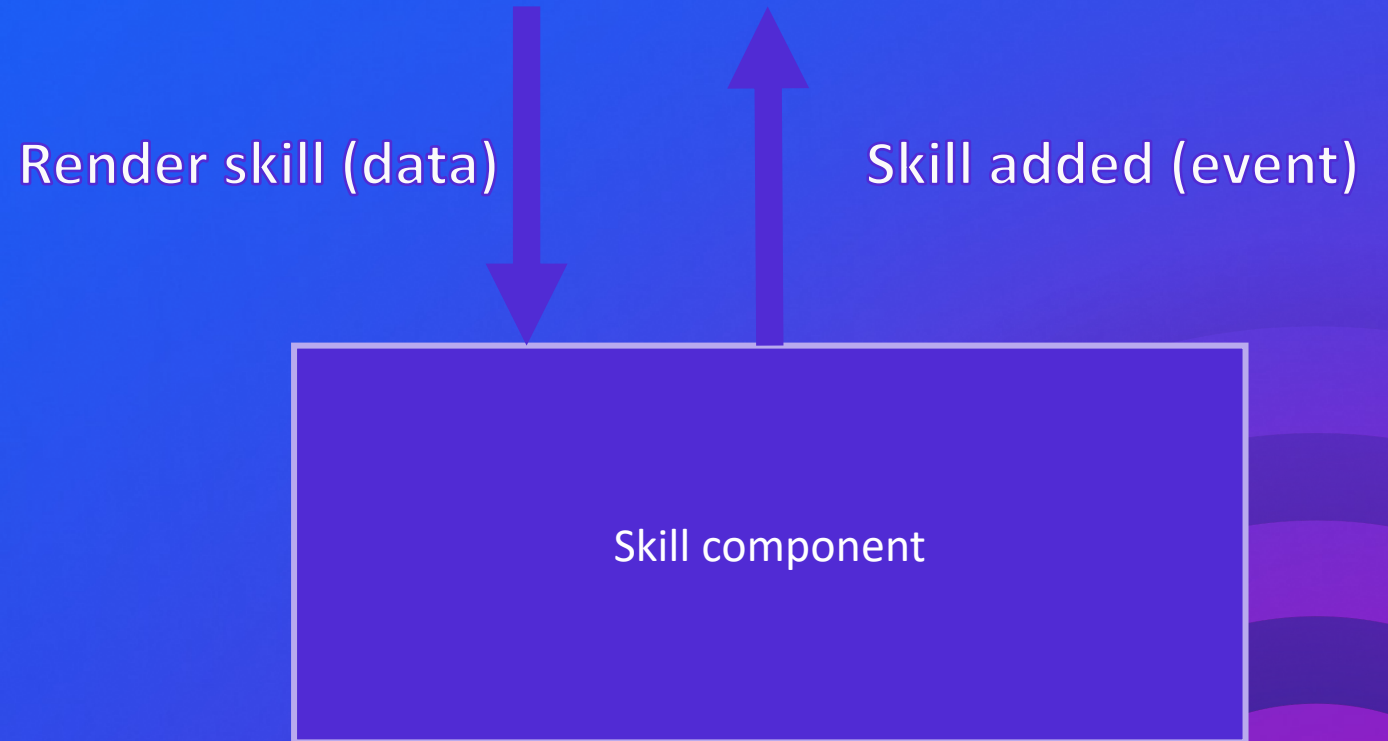
A components is something that should do one thing well

Comes in two different flavours

- Presentation components, "just" presents the data
- "Smart" components, manages data and events

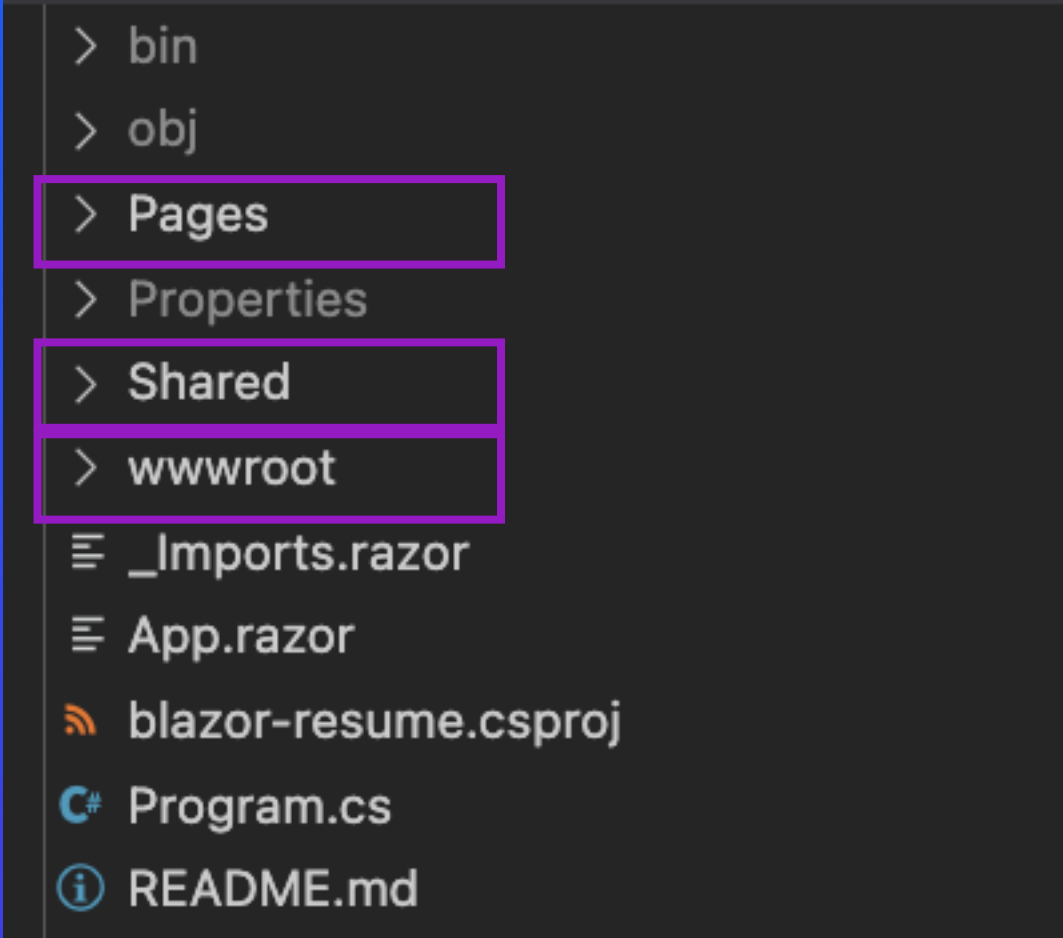
Architectural principles

Data flows down
Events flow up



Create a blazor app

```
dotnet new blazorwasm -o  
ResumeApp
```



- > bin
- > obj
- > **Pages**
- > Properties
- > **Shared**
- > **wwwroot**
- ≡ _Imports.razor
- ≡ App.razor
- 📄 blazor-resume.csproj
- 📄 Program.cs
- 📄 README.md

A component in Blazor

- Navigation with @page
- Render data with @
- Respond to events with @event="EventHandler"

```
@page "/counter"
```

```
<PageTitle>Counter</PageTitle>
```

```
<h1>Counter</h1>
```

```
<p role="status">Current count: @currentCount</p>
```

```
<button class="btn btn-primary" @onclick="IncrementCount">Click  
me</button>
```

```
@code {
```

```
private int currentCount = 0;
```

```
private void IncrementCount()
```

```
{
```

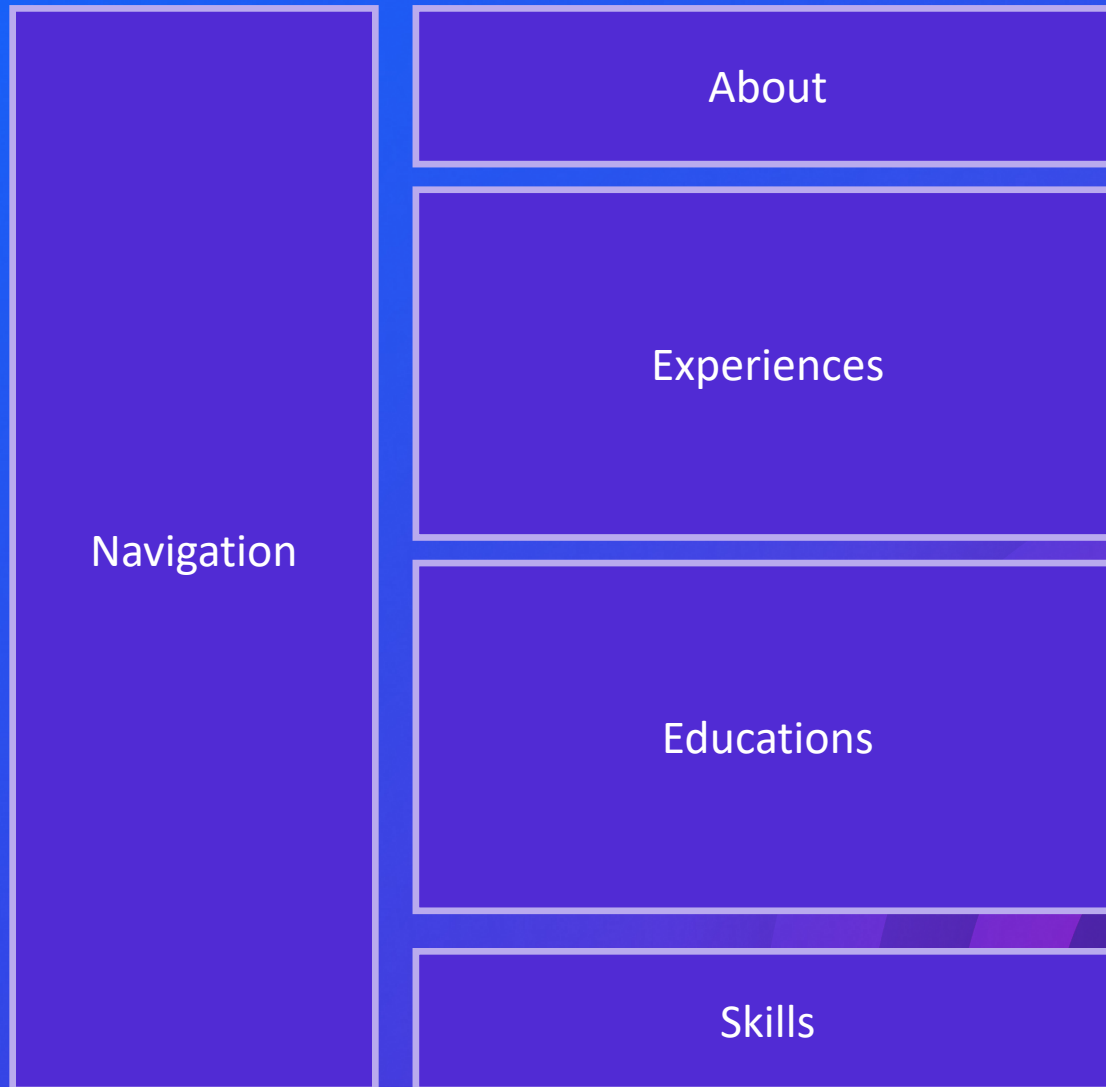
```
currentCount++;
```

```
}
```

```
}
```

Architecting our app

Components



Resume - navigation

Shared

≡ MainLayout.razor

MainLayout.razor.css

≡ NavMenu.razor

```
<div class="nav-item px-3">  
<NavLink class="nav-link" href="resume">  
<span class="oi oi-plus" aria-hidden="true"></span> Resume  
</NavLink>  
</div>
```

Resume – resume component

- Setup navigation @page, /resume
- Define “name” and render it in template area

```
@page "/resume"
```

```
<PageTitle>@name</PageTitle>
```

```
<div class="about">  
<h2>About</h2>  
</div>
```

```
@code {  
    string name = "matt";  
}
```

What does our JSON data look like?

```
✓ wwwroot
  > css
    ✓ sample-data
      {} resume.json
```

```
{
  "about" : ".NET developer with a proven track
record and with a growth mindset. Additional
skills in HTML and JavaScript and CSS",
  "experiences" : [{
    "companyname": "Microsoft",
    "title" : "Software developer",
    "tenure" : "2020-2022",
    "description" : "Software developer. As part
of a team, built 2 services that served 20 000
users and improved the service's profit by 10%"
  },
  ...
  ],
  "educations": [ ...
  ],
  "skills" : [{ ...
  },
  ...
  ],
}
```

How do we model that in C#?

```
using System.Collections.Generic;
namespace Types;

public class ResumeModel {
    public string About { get; set; }

    public List<ExperienceModel> Experiences { get; set; }
    public List<EducationModel> Educations { get; set; }
    public List<SkillModel> Skills { get; set; }
}

public class ExperienceModel
{
    public string CompanyName { get; set; }
    public string Tenure { get; set; }

    public string Description { get; set; }
    public string Title { get; set; }
}
```

```
public class EducationModel
{
    public string Name { get; set; }
    public string DegreeYear { get; set; }
    public string Description { get; set; }
}

public class SkillModel
{
    public string ? Name { get; set; }
    public int ? EstimatedLevel { get; set; }
    public string ? Description { get; set; }
}
```

Resume – resume component, fetch data

- Inject HTTP capability
- Define OnInitializedAsync()
- Request data from JSON file

```
@page "/resume"
@using Types
@Inject HttpClient Http

<PageTitle>@name</PageTitle>

<div class="about">
<h2>About</h2>
@resume.About
</div>

@code {
    string name = "matt";
    public ResumeModel resume { get; set; }

    protected override async Task OnInitializedAsync()
    {
        resume = await Http.GetFromJsonAsync<ResumeModel>("sample-
data/resume.json");
    }
}
```

Let's render the fetched data

```
<div class="experiences">
<h2>Experiences</h2>
@foreach (var experienceModel in resume.Experiences)
{
<Experience data=@experienceModel></Experience>
}
</div>
```

```
<div class="educations">
<h2>Education</h2>
@foreach (var educationModel in resume.Educations)
{
<Education data=@educationModel></Education>
}
</div>
```

```
<div class="skills">
<h2>Skills</h2>
@foreach (var skillModel in resume.Skills)
{
<Skill data=@skillModel></Skill>
}
</div>
```

Resume – Educations, Skills, Experiences

```
@using Types
```

```
<div class="experience">  
<h3>@data?.Title</h3>  
<h4>@data?.CompanyName (@data?.Tenure)</h4>  
<div>@data?.Description</div>  
</div>
```

```
@code {
```

```
[Parameter]
```

```
public ExperienceModel? data { get; set; }  
}
```

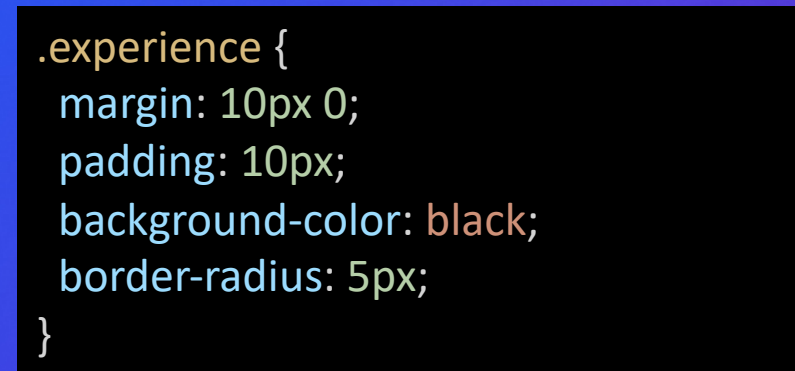
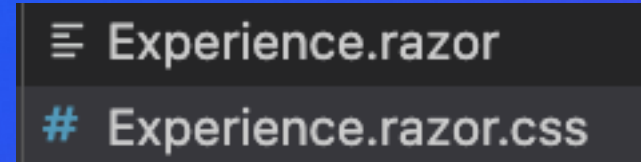
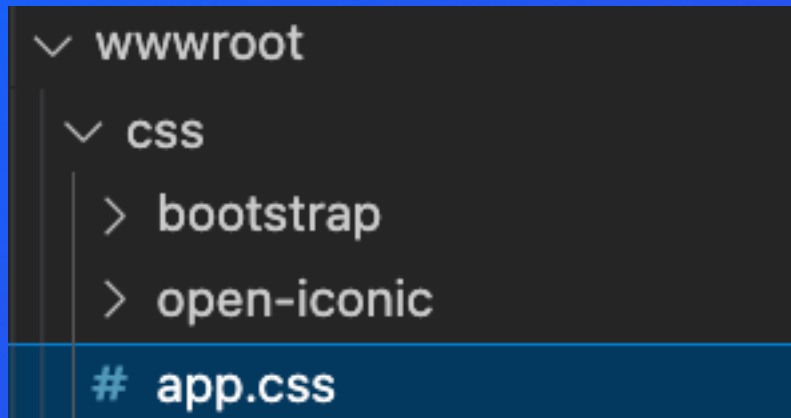
Resume – styling, close to the component

Styling should be close to the component

Component.razor

Component.razor.css

Some exception where you add CSS app wide in





DEMO

Student Resources

<http://aka.ms/learnstudent>

Cloud Skills Challenge

<https://aka.ms/dotnetstudentcsc>

GitHub Repo

<https://github.com/microsoft/dotnetconf-studentzone>